New Interactive Visual Tools and Statistical Methodology for Selecting and Evaluating Nonlinear Dimension Reduction Layouts of High-Dimensional Data

Response to reviews

We thank the examiners for their careful assessment of my thesis. What follows is a point by point response to the examiners' comments.

Examiner A: Prof Christian Hennig

The thesis treats the evaluation of two-dimensional representations of high-dimensional data by nonlinear approaches such as UMAP and t-SNE. The core is the use of hexagonal binning of the 2-d representation, which can be lifted into the higher dimensional space, used to visualise the relation between the 2-d representation and the original data using a tour, and to assess the quality of the fit by the so-called hexbin error (HBE). This can in particular be used to compare different representations. The thesis presents various illustrations and visualisations based on this approach. Chapter 2 presents the basic definitions and methods. Chapter 3 presents a software package to implement these. Chapter 4 presents another package that can generate benchmark datasets from an impressive amount of different models generating geometrically structured data. This can in particular be used to generate data with different clusters, potentially of very different kinds. Chapter 5 contains an empirical study in which test persons were asked to guess whether a low-dimensional representation is of the same dataset that they could see in a tour. In particular it was investigated in which way this depends on separation between clusters for a variety of representation models. Chapter 6 provides a Shiny App that allows a user to use the techniques proposed in Chapter 2 to assess different representations of their own data in an interactive manner. Chapter 7 concludes the thesis.

The thesis is very well written. The material presented is original, useful, timely, and of wide interest. As far as I see, it is also correct. The software seems well organised, well documented, and well thought through.

As the work is collaborative, I am not in a position to assess the candidate's individual contributions precisely, however I will assume that the candidate did much of the work and is willing to take responsibility for all of it.

Assuming this, I confirm that

- the thesis makes an original and substantial contribution to the discipline or area of professional practice;
- the candidate is able to critically reflect on, and engage with, complex ideas to create new knowledge and understanding;
- the candidate has presented a thesis which, in format and presentation, is appropriate to the standard expected of the degree.

I have a number of requests for amendments. In particular, the work touches upon some rather profound issues that the candidate currently treats rather naively, and that deserve more acknowledgment and discussion.

Therefore I recommend a pass with minor edits required. Chances are the edits can be made in 1-2 months. What I write below is quite long, but much of this is just to explain the issue. I'm asking for acknowledgment and some thoughts regarding these issues, not for any rewriting, much change of what is there, or much extra work.

Major issues:

General The HBE is used as a general metric in order to evaluate representations, and it is stated in several places that a representation with a better HBE "performs better". This is far from clear. In fact there are many statistics around that can be interpreted as evaluating the quality or lower dimensional representations. Actually many methods providing such representations are based on optimising an objective function that is meant to measure representation quality in some sense, starting from the variance criterion in PCA, various versions of stress in MDS, trustworthiness etc. Even t-SNE and UMAP can be written down as optimisation problems. Section 2.4 mentions and uses a number of other metrics, but formal definitions are not given, and there is no discussion regarding the pros and cons of these metrics. Given that this is so, why could HBE be expected to formalise the quality of a representation any better than the metrics that already exist? Actually HBE has the rather obvious disadvantage that it depends on the parameters of the grid, which is not the case for most other existing metrics. So what is its actual advantage? Before recommending a software to the user that assesses "performance" by means of HBE, this surely needs to be addressed.

General In many places the aim seems to be to find the "best" representation. But data analysts often use many visualisations of the same data because this can bring additional insight. Different kinds of visualisation can highlight different features of the data, and most insight can come from combining what can be learnt from several displays. So it should at least be acknowledged that bringing together a number of representations could be more informative for data analysis than looking for a single "best" one.

General Cluster analysis is quite prominent in the thesis, and it is apparently implicitly always assumed that every dataset has a uniquely true clustering. There are several references to the "true clusters" without ever defining what these are supposed to be. This is very naive. There is no unique and agreed definition of what a cluster is, and several different legitimate clusterings can exist in the same dataset. See in particular

Christian Hennig (2015) What are the true clusters? Pattern Recognition Letters 64, 53-62, https://doi.org/10.1016/j.patrec.2015.04.009.

So if there is talk of "true clusters", there would need to be a definition of what this means, either (preferably) as functional of a general probability distribution, or using a data-based definition.

The implicit assumption that the true clusters are unique and clear is particularly implied in Section 4 where different "shapes" of clusters can be combined in a single dataset. I'd like to challenge the author to give a general definition of true clusters according to which the true clusters are indeed in all these situations those that are implied to be the true cluster here, but I'm afraid chances are that this would be rather hopeless. Appeal to intuition will not do, and neither will be a statement that "different clusters are generated by different distributions" because any mixture distribution can be split up in an infinite variety of ways into different distributions. So the best I can ask for is an informal but insightful discussion of the issue, including a clear acknowledgment that the thesis uses an informal and intuitive cluster concept that can in all likelihood not be made precise and may very well involve ambiguity. There is more than one legitimate clustering of a dataset, so it cannot be fully clear what the "true clusters" are that are mentioned.

The issue is particularly relevant when benchmarking clustering methods (Sec. 4.4.2). All results are relative to the assumed truth. Note that methods like kmeans can be seen as coming with their own implicit cluster concept, and they will deviate obviously and legitimately from clusters that are defined in a different way, as are the clusters here.

General In examples, PCA is usually run on high-dimensional data to reduce the dimension before applying the methods of interest to then only reduce the 10 (or so)-dimensional PCA representation to two dimensions. There should be a comment on this, why this is done and whether this is generally recommended. As far as I know, the literature on UMAP and other techniques treated here doesn't recommend this and doesn't suggest that this would be necessary.

Sec. 4.4.2 Much about this Section is quite naive and not very insightful. References for standard cluster analysis methods that are around for 50 years and more are given from between 2002 and 2019; original references are preferred. There are lots of hierarchical methods and it is not specified which one is used here. Also the concept of model-based clustering is very general and although I can guess what the author did, the given description is insufficient (not only Gaussian distributions can be mixed to define a clustering model). Given the variety of available covariance matrix models it seems very strange that the authors picked the one model that is almost identical to k-means when comparing with k-means. This is hardly ever of interest in practice, because otherwise one could stick to k-means in the first place, and the superior flexibility of model-based clustering would be wasted.

A number of cluster validity statistics are computed, but there isn't much discussion of their meaning and whether or why some of these might be more appropriate than others. In fact some of these are not even meant to be used on their own, see

Akhanli S.E., Hennig C. (2020). Comparing clusterings and numbers of clusters by aggregation of calibrated clustering validity indexes. STATISTICS AND COMPUTING, 30(5), 1523-1544, doi:10.1007/s11222-020-09958-2

Some guidance on good practices for benchmarking in cluster analysis:

Van Mechelen, I., Boulesteix, A.-L., Dangl, R., Dean, N., Hennig, C., Leisch, F., Steinley, D., & Warrens, M. J. (2023). A white paper on good research practices in benchmarking: The case of cluster analysis. WIREs Data Mining and Knowledge Discovery, 13(6), e1511. https://doi.org/10.1002/widm.1511

Chapter 5 There are some problematic issues with this survey. As far as I understand, many high-dimensional datasets can lead to the same 2-d representation using the techniques discussed here. This means that participants can at best know that it is possible that datasets are the same (2-d representation and tour), but it is always just as well possible that datasets are different. Even with the best understanding of the methods and the best intuition this can never be excluded. Saying "same" with confidence is therefore always wrong, because one can guess "same", but confidence is never justified. Furthermore, I'd think that for the use of techniques such as UMAP, t-SNE etc. some understanding and training is required. They are not necessarily meant (or at least shouldn't be meant) to be used in a purely intuitive way without any idea of their inner workings. But as far as I understand, this kind of intuitive use is what the survey investigates. There should be some discussion of to what extent this is appropriate and informative.

Minor issues:

p. 6 and elsewhere "NLDR can hallucinate wildly" - without a formal definition of what true structures are and how they should or should not be represented, it isn't clear what "hallucinate" means. Also arguably the methods just do what they are defined to do; without a human analyser misinterpreting the display there can be no "hallucination".

p. 8 Scale the data - I worry that there may be robustness issues with the min/max scaling here. A single large outlier on a variable will imply that all other observations have hardly any distance between each other on this variable.

p. 9 Write the linear optimisation problem down explicitly. This is not transparent.

p.11 What does "RNX curve" mean?

Fig. 2.5 y-axis label and annotation are missing.

p. 16, l. -6 "high correlation value" between what and what?

Fig. 2.8 bottom The difference between the different metrics in assessing the layouts is staggering. This deserves a discussion.

Fig. 2.12 Layout f is called "best choice" but it doesn't have the best HBE values where the HBE values are best (at low average bin count). What is the rationale here?

p. 59 More published data generators for clustering are:

Maitra, R., & Melnykov, V. (2010). Simulating Data to Study Performance of Finite Mixture Modeling and Clustering Algorithms. Journal of Computational and Graphical Statistics, 19(2), 354–376. https://doi.org/10.1198/jcgs.2009.08054

C. Shand, R. Allmendinger, J. Handl, A. Webb and J. Keane, "HAWKS: Evolving Challenging Benchmark Sets for Cluster Analysis," in IEEE Transactions on Evolutionary Computation, 26 (6), 1206-1220, 2022, doi: 10.1109/TEVC.2021.3137369

p. 62 I haven't exactly understood how the "background noise" works. Better explain such things formally and unambiguously.

p. 64 "they do not produce a formal multicluster dataset" - what is that?

By "formal multicluster dataset", we mean a dataset composed of multiple distinct and separable groups (clusters), where each cluster represents an independent structure, typically with clear boundaries or low-density separation between them.

In contrast, the branching structures generated here form a single connected object with multiple arms, so they do not consist of independent clusters despite having visually distinct components.

p. 68, l.-9 There's a bracket too many.

We removed the additional bracket.

p. 70, l.-9 Explain "Sierpinski-like".

By "Sierpinski-like", we refer to a structure created by repeatedly taking midpoints between points and the corners of a shape. This process produces a pattern with a repeating structure and visible gaps, similar to the well-known Sierpiński triangle and its higher-dimensional versions.

We use the term "-like" to indicate that the dataset only approximates this pattern using a finite number of points, rather than forming an exact mathematical fractal.

Chapter 5 I think results for PCA should be included. They would certainly be of interest, and not doing them because of some speculation what the result may be isn't convincing.

p. 91 BW ratio treats the clusters as spherical, not just elliptical.

We agree that the between–within (BW) ratio is most naturally suited to spherical cluster assumptions, rather than general elliptical shapes. We revised the text as: Measuring distance between clusters is classically done using the between–within (BW) ratio, which captures global separability under assumptions of roughly spherical cluster structure.

p. 92 Explicitly introduce the terminology “SAME” and “DIFFERENT” trials. Don’t leave guesswork to the reader, not even if names are expressive.

We added the text as: The experiment consists of two types of trials: SAME trials and DIFFERENT trials. In SAME trials, two visualizations (one NLDR layout and the tour) are generated from the same underlying dataset, but with controlled variations in cluster separation. In DIFFERENT trials, the two visualizations are generated from different datasets.

p. 92 What qualifies data to be “attention check data”? Gaussian variance is specified as a single number, but in order to define the Gaussian’s properly you need to state their covariance matrices.

p. 95 “The BW-ratio. . .” is not a sentence.

Sec. 5.3.7 I think that once the same dataset (or generation rule) has been used more than once, the model needs a dataset effect, probably another random effect. Results of different test persons on the same data are dependent for sure.

Fig. 5.4 and elsewhere Estimated probability results look generally quite weak given that there are only two response options. Comment!

Fig. B.10b How is what is shown here connected to normal distribution of the random effects? The assumption is not about the plain “correct proportion”.

B.8.1 “Subjects who exceeded the average time of 5-10 minutes were excluded, as determined from the pilot study.” Why? I don’t see any reason for this.

“Finally, the collected data set is further refined by filtering out all the responses which showed the same data structures in 2-D NLDR plot and tour.” I suspect this sentence doesn’t say what you want to say here. In any case it is very strange.

Examiner B: Prof Anne M. Ruiz

No comments.